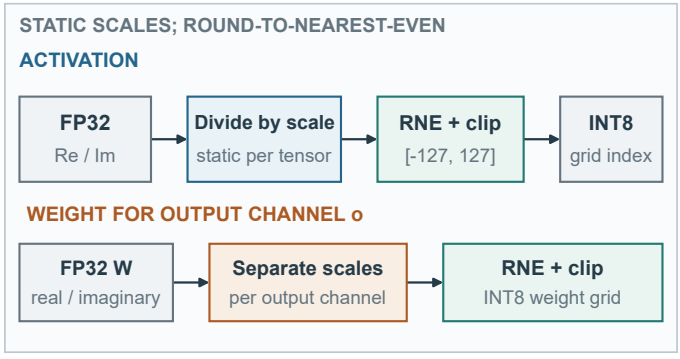
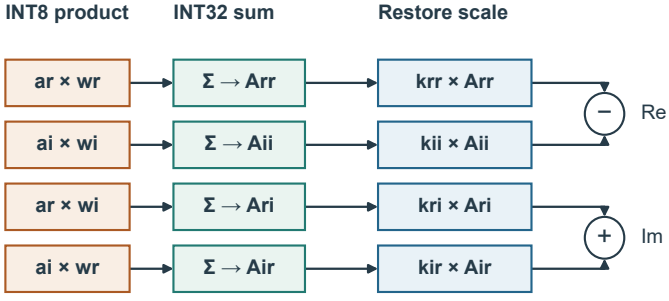
a Signed grids and separately calibrated scales



Activation scale: 1.02 × observed maximum / 127

b Four integer sums; four scale-restoration ratios

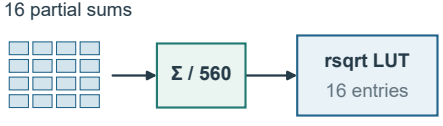


Combine scaled sums → one RNE + clip per output boundary

Unequal Re / Im scales prohibit a common unscaled complex sum.

c Mixed-precision numerical helpers implemented in QAT and HLS

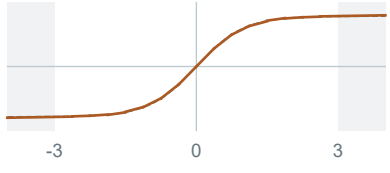
PER-SLOT RMS NORMALIZATION



Exponent factor restores the reciprocal scale

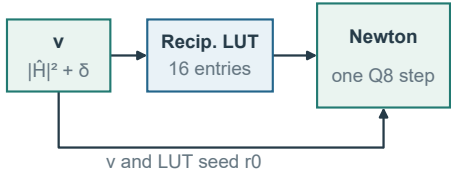
No Newton step; selected helpers retain FP32

EXPORTED 16-segment tanh



Solid: exported surrogate; dashed: tanh

RECIPROCAL + NEWTON CORRECTION



Output: $Y\hat{H}^* \cdot r$; then layer requantization

FFT/IFFT and selected numerical helpers remain floating point.